

ADASOLVER: EFFICIENT NEURAL OPERATOR INFERENCE VIA TEST-TIME ADAPTIVE SAMPLING

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ABSTRACT

Neural operators provide efficient surrogates for partial differential equations (PDEs), but their inference cost is typically tied to a fixed spatial discretization or uniform subsampling. Such discretizations can waste computation when fine resolution is needed only in localized regions. However, changing the sampling density shifts the empirical measure in latent aggregation and can alter the continuous operator represented by a frozen point-based model. We introduce **AdaSolver**, a measure-aware framework for adaptive spatial computation at test time. We characterize sampling-induced bias and its propagation through Physics-Attention, then derive a sampling-density-consistent Transolver using self-normalized importance weighting. The corrected aggregation remains consistent with the reference measure, enabling adaptive point allocation guided by geometry and physics priors, coarse predictions, or a cache-aware learned refinement policy. We further extend variable-resolution learning to distribution-amortized training over both point counts and sampling densities. Experiments on five standard PDE benchmarks and three industrial aerodynamic prediction tasks show improved accuracy at matched point budgets, with solution-informed policies providing the strongest gains on the standard benchmarks. Airfoil and wing optimization further demonstrate the value of adaptive sampling for accelerating surrogate-based design. These results establish spatial sampling as a principled axis of test-time scaling for neural operators.

1 INTRODUCTION

Neural operators learn maps between function spaces and can amortize expensive numerical simulation across many physical configurations (Li et al., 2021; 2023c; Wu et al., 2024). Recently, transformer-based neural operators have improved the scalability and accuracy of operator learning across diverse PDEs and million-point geometries. They make such large-scale computation possible by compressing irregular meshes into latent tokens, distributing the computation, or decoupling global state estimation from pointwise decoding (Alkin et al., 2024; Luo et al., 2025; Zhou et al., 2026). In engineering applications, however, evaluation efficiency is as important as the ability to process large inputs. Yet how neural operators can flexibly allocate computation at inference remains largely unexplored.

Neural-operator training data are typically generated by numerical solvers on meshes that are uniform or designed for numerical convergence. Learned models are consequently evaluated under the same sampling distribution, which may ignore the strongly localized structure of many PDE solutions. Shocks, boundary layers, contact interfaces, and stress concentrations often occupy only a small part of the domain and require finer local resolution, whereas smooth regions can be represented with fewer samples. This creates a dilemma: preserving the training-time distribution wastes points in smooth regions, while reallocating them toward critical regions changes the input sampling distribution. Can a neural operator redistribute a fixed spatial budget without changing the operator it represents?

Existing efforts address neighboring capabilities. First, resolution generalization studies how neural operators extrapolate across grid sizes, point counts, or compatible discretizations while preserving the underlying sampling distribution (Li et al., 2021; 2023c; Liu et al., 2023; Alkin et al., 2024; Zhou et al., 2026; Gao et al., 2025). These methods make resolution changes feasible, but do not by

themselves provide a mechanism for actively changing the local sampling density. Second, adaptive sampling and refinement methods predict where a numerical mesh or point set should be refined, often as part of a solver pipeline or through a separately trained refinement model (Huang et al., 2021; Obiols-Sales et al., 2023; Rowbottom et al., 2024). They address spatial allocation, but do not specify how a frozen neural operator should preserve its target integral when that allocation changes the sampling measure. Third, test-time compute scaling increases inference effort through adaptive reasoning or iterative physical prediction (Snell et al., 2024; Lippe et al., 2023; Jenkins, 2026; Serrano et al., 2026). These works establish the value of allocating additional computation at inference, but leave open the spatial counterpart: reallocating input samples for operator evaluation while retaining the same learned map.

We introduce AdaSolver to address this gap. We view spatial resolution, controlled by the input sampling density, as a controllable inference resource. From a continuous-operator perspective, we characterize how the discrepancy between the inference-time sampling density and the reference distribution biases both the latent physical states and the predicted output function. Based on this analysis, we develop a sampling-density-consistent Transolver that uses self-normalized importance weighting to correct the sampling bias in physics-slice aggregation. This correction enables the model to construct an adaptive inference density from geometry or physics priors, a low-cost draft prediction, or a lightweight head attached to reusable intermediate features. We further incorporate these sampling strategies into training, supporting both geometry-amortized training across point sets and distribution-amortized training across sampling densities.

Our contributions are threefold:

- We formulate sampling density as the mechanism for controlling spatial test-time compute and derive how its deviation from the reference distribution biases the latent physical states and output function of a transformer neural operator.
- We propose a sampling-density-consistent Transolver that corrects physics-slice aggregation through self-normalized importance weighting, preserving the reference operator under inference-time density changes.
- We develop prior-guided, draft-guided, and cache-aware learned sampling strategies, and incorporate them into geometry-amortized and distribution-amortized training.

2 RELATED WORK

Neural operators on irregular geometries. Fourier neural operators learn resolution-transferable spectral mappings on regular grids, while Geo-FNO extends them to general geometries through learned coordinate deformations (Li et al., 2021; 2023b). GINO couples integral operators with a latent regular grid, and NUNO partitions non-uniform observations to reduce interpolation and computational costs (Li et al., 2023c; Liu et al., 2023). Transformer-based operators instead compress large point sets into latent representations before global interaction, as in GNOT, UPT, and Transolver (Hao et al., 2023; Alkin et al., 2024; Wu et al., 2024). Transolver++ and Transolver-3 further scale this design to million-scale and industrial-scale geometries (Luo et al., 2025; Zhou et al., 2026). These methods establish strong compatibility with irregular inputs and changing point counts. Our focus is complementary: we actively change the *local* sampling density at inference and ask whether the frozen model still represents the same continuous operator.

Discretization-consistent operator learning. Extending finite neural architectures to function spaces requires discrete reductions to converge to fixed continuous integrals as the discretization changes (Berner et al., 2026). Monte Carlo convolution corrects non-uniform point-cloud sampling with inverse-density weights, while QuadConv evaluates continuous kernels using quadrature on irregular PDE data (Hermosilla et al., 2018; Doherty et al., 2024). Related formulations construct discretization-invariant maps between neural fields and quadrature-aware normalization statistics (Wang & Golland, 2022; Kim et al., 2026). Nevertheless, resolution-independent parameters alone do not prevent errors under a mismatch between training and evaluation discretizations (Gao et al., 2025). We adopt the same function-space principle for latent point aggregation, but use it to support a deliberate inference-time change of sampling density rather than only passive resolution transfer.

Adaptive meshes and test-time refinement. Adaptive discretization concentrates resolution where numerical error is expected to be large. Learned approaches predict mesh density, refinement, or node relocation for numerical solvers (Huang et al., 2021; Obiols-Sales et al., 2023; Rowbottom et al., 2024), while MeshGraphNets integrates learned dynamics with mesh-based simulation (Pfaff et al., 2021). Test-time computation has also been increased through iterative denoising in PDE-Refiner, adaptive allocation or tokenization, and neural-operator composition (Lippe et al., 2023; Jenkins, 2026; Wang & Perdikaris, 2026; Serrano et al., 2026). These directions determine where or how additional inference computation should be used. AdaSolver addresses the complementary operator-level requirement: changing the spatial sampling density of a frozen point-based neural operator while keeping its reference continuous measure fixed.

3 METHOD

3.1 TRANSFORMER NEURAL OPERATORS

Transformer neural operators use attention to approximate nonlocal integral operators. Let Ω denote the physical domain, and let ρ denote the sampling measure induced by the chosen grid or mesh discretization policy. A discretized example consists of pointwise samples $\{(x_i, a_i)\}_{i=1}^N \mapsto \{u_i\}_{i=1}^N$, where $x_i \sim \rho$, $a_i = a(x_i)$ is the input field, and $u_i = u(x_i)$ is the output field. A neural operator learns an operator $\mathcal{G} : a \mapsto u$ with a parameterized network $\mathcal{G}_\theta$.

OFormer (Li et al., 2023a) uses Galerkin attention, a softmax-free linear-attention factorization that reduces the quadratic point-to-point interaction to linear dependence on N for fixed channel width. After pointwise projections, let $Q, K \in \mathbb{R}^{N \times d}$ and $V \in \mathbb{R}^{N \times C}$, where d is the attention width and C is the number of value channels. Denote the ℓ -th query and key basis functions by q_ℓ and k_ℓ , and the c -th value basis function by v_c ; their pointwise evaluations are obtained from the embedded samples (x_i, a_i) . With column-wise normalized quantities denoted by a hat, Galerkin attention is

$$y_{i,c}^{(N)} = \sum_{\ell=1}^d \hat{q}_\ell(x_i) \left(\frac{1}{N} \sum_{n=1}^N \hat{k}_\ell(x_n) \hat{v}_c(x_n) \right) \xrightarrow{N \rightarrow \infty} \sum_{\ell=1}^d q_\ell(x_i) \int_{\Omega} k_\ell(x) v_c(x) d\rho(x). \quad (1)$$

Here q_ℓ and k_ℓ span the learned query and key bases, while v_c denotes a value channel. Equivalently, the update is $Y = \hat{Q}(\hat{K}^\top \hat{V})/N$, which directly approximates the corresponding integral operator.

Transolver (Wu et al., 2024) replaces point-to-point attention with Physics-Attention: point features are projected into M slice weights, attention is applied to the resulting latent physical states, and the updated states are broadcast back to the points. To avoid overloading the sample locations x_i , let $h_i \in \mathbb{R}^C$ denote the embedded point feature obtained from (x_i, a_i) . The slice weights are

$$w_i = \text{Softmax}\left(\frac{\text{Linear}(h_i)}{\tau_0}\right) \in \mathbb{R}^M, \quad \sum_{j=1}^M w_{i,j} = 1, \quad (2)$$

where τ_0 is the temperature. The corresponding physics token is the normalized weighted aggregation

$$u_{s,j}^{(N)} = \frac{\frac{1}{N} \sum_{i=1}^N w_{i,j} h_i}{\frac{1}{N} \sum_{i=1}^N w_{i,j}} \xrightarrow{N \rightarrow \infty} u_{s,j}^\rho = \frac{\int_{\Omega} w_j(x) h(x) d\rho(x)}{\int_{\Omega} w_j(x) d\rho(x)}. \quad (3)$$

Writing $u_s^{(N)} = [u_{s,1}^{(N)}, \dots, u_{s,M}^{(N)}]^\top \in \mathbb{R}^{M \times C}$ and $q, k, v = \text{Linear}(u_s^{(N)})$, the discrete state attention and its continuous integral counterpart can be written together as

$$\begin{aligned} u'_{s,j} &= \sum_{t=1}^M \frac{\exp(q_j^\top k_t / \tau)}{\sum_{p=1}^M \exp(q_j^\top k_p / \tau)} v_t \\ &\xrightarrow{M \rightarrow \infty} \mathcal{A}[u_s^\rho](\xi'_s) \\ &:= \int_{\Omega_s} \frac{\exp(q(u_s^\rho(\xi'_s))^\top k(u_s^\rho(\xi_s)) / \tau)}{\int_{\Omega_s} \exp(q(u_s^\rho(\xi'_s))^\top k(u_s^\rho(\eta_s)) / \tau) d\eta_s} v(u_s^\rho(\xi_s)) d\xi_s. \end{aligned} \quad (4)$$

where $u_s^\rho(\xi_s)$ is the continuous physics-token field on the slice domain Ω_s , ξ'_s is a query location, and τ is the attention temperature. Finally, the updated tokens are mapped back to the point space through the slice weights,

$$h'_i = \sum_{j=1}^M w_{i,j} u'_{s,j}, \quad i = 1, \dots, N. \quad (5)$$

Thus, Physics-Attention replaces the $N \times N$ point interaction with operations of complexity $\mathcal{O}(NMC + M^2C)$ for $M \ll N$. During training, the point sets used by $\mathcal{G}_\theta$ are typically generated according to the prescribed measure ρ .

3.2 SAMPLING-DENSITY-CONSISTENT TRANSOLVER

A decoder-only transformer neural operator is typically trained and evaluated on point sets generated according to a prescribed sampling measure ρ , which is determined by the discretization used for data generation, such as a uniform grid or a geometry-dependent mesh-refinement policy. At inference time, we may instead sample from a different measure $\tilde{\rho}$ to obtain a different spatial description of the same physical domain. For example, aerodynamic geometries can be sampled more densely near surfaces or flow-sensitive structures and more sparsely in smooth exterior regions. However, because transformer neural operators approximate continuous operators through Monte Carlo integration, changing the sampling measure generally changes the limiting operator represented by the frozen model.

Let $\lambda = d\tilde{\rho}/d\rho > 0$ denote the density ratio. For a slice location $\xi_s \in \Omega_s$, define the slice-weighted probability measure

$$d\pi_{\xi_s}(x) := \frac{w(x, \xi_s)}{\int_{\Omega} w(y, \xi_s) d\rho(y)} d\rho(x). \quad (6)$$

Proposition 3.1 (Physics-token bias under a sampling-density shift). *For each $\xi_s \in \Omega_s$, the reference and shifted physics tokens satisfy*

$$u_s^{\tilde{\rho}}(\xi_s) - u_s^\rho(\xi_s) = \frac{\text{Cov}_{\pi_{\xi_s}}(\lambda, h)}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]}. \quad (7)$$

Moreover, if $\mathcal{P}_{\xi_s}^\mu h := u_s^\mu(\xi_s)$ denotes the tokenization map under a sampling measure μ , then

$$\left\| \mathcal{P}_{\xi_s}^{\tilde{\rho}} - \mathcal{P}_{\xi_s}^\rho \right\|_{\text{op}} = \frac{\sqrt{\text{Var}_{\pi_{\xi_s}}(\lambda)}}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]}. \quad (8)$$

Let $\mathcal{T}$ denote the continuous Physics-Attention operator and define $\mathcal{G}_\mu(h) := \mathcal{T}[u_s^\mu]$. Write $\delta_s := u_s^{\tilde{\rho}} - u_s^\rho$ and let $R_{\delta_s} := \alpha_{u_s^{\tilde{\rho}}} / \alpha_{u_s^\rho}$ be the ratio between the shifted and reference attention kernels.

Theorem 3.2 (Propagation through continuous Physics-Attention). *The sampling-density bias propagates to the Physics-Attention output as*

$$\begin{aligned} \mathcal{G}_{\tilde{\rho}}(h)(x) - \mathcal{G}_\rho(h)(x) &= \int_{\Omega_s} w(x, \xi'_s) \int_{\Omega_s} \alpha_{u_s^\rho}(\xi'_s, \xi_s) \left[(R_{\delta_s}(\xi'_s, \xi_s) - 1) W_v u_s^\rho(\xi_s) \right. \\ &\quad \left. + R_{\delta_s}(\xi'_s, \xi_s) W_v \delta_s(\xi_s) \right] J_g(\xi_s) d\xi_s d\xi'_s. \end{aligned} \quad (9)$$

Here $\alpha_{u_s^\rho}$ is the reference attention kernel, W_v is the value projection, and J_g is the physical-to-slice Jacobian. The two terms respectively describe the redistribution of attention weights and the propagation of shifted value tokens.

The complete conditions and proofs of Proposition 3.1 and Theorem 3.2 are given in Appendix A.

These results identify the point-to-slice aggregation as the source of the sampling-density bias. The slice assignment $w_{i,j}$ is computed independently at each point, attention is subsequently applied among a fixed number of physics tokens, and deslicing maps the updated tokens back to each point. Only the physics-token construction aggregates over the sampled points. Consequently, for $x_i \sim \tilde{\rho}$, the original aggregation converges to

$$\frac{\sum_{i=1}^N w_{i,j} h_i}{\sum_{i=1}^N w_{i,j}} \xrightarrow{N \rightarrow \infty} \frac{\int_{\Omega} w_j(x) h(x) d\tilde{\rho}(x)}{\int_{\Omega} w_j(x) d\tilde{\rho}(x)} = u_{s,j}^{\tilde{\rho}}, \quad (10)$$

rather than the reference token $u_{s,j}^\rho$ learned under ρ .

We therefore correct the measure only when forming the physics tokens. For each inference point, let

$$c_i := \frac{d\rho}{d\lambda}(x_i) = \lambda(x_i)^{-1}, \quad (11)$$

be its importance weight, or the corresponding inverse relative inclusion density for discrete sampling. The corrected aggregation is

$$\hat{u}_{s,j}^{(N)} := \frac{\sum_{i=1}^N c_i w_{i,j} h_i}{\sum_{i=1}^N c_i w_{i,j}} \xrightarrow{N \rightarrow \infty} \frac{\int_{\Omega} w_j(x) h(x) d\rho(x)}{\int_{\Omega} w_j(x) d\rho(x)} = u_{s,j}^\rho. \quad (12)$$

Because the estimator is self-normalized, only the relative density ratios are required. For finite samples, the implementation may replace c_i with c_i^β , where $\beta = 0$ recovers the original Transolver and $\beta = 1$ gives the measure-consistent correction above.

After the corrected physics tokens are obtained, we apply the original attention among physics tokens and the original deslice operation without further modification. For a Transolver with L layers, the same correction is applied whenever layer ℓ forms new physics tokens from its current point features $h_i^{(\ell)}$:

$$\hat{u}_{s,j}^{(\ell)} = \frac{\sum_{i=1}^N c_i w_{i,j}^{(\ell)} h_i^{(\ell)}}{\sum_{i=1}^N c_i w_{i,j}^{(\ell)}}, \quad \ell = 1, \dots, L. \quad (13)$$

Each layer then continues with its unchanged Physics-Attention and deslice operations. Thus, the correction follows the point-to-slice aggregation throughout the frozen network while preserving all learned projections and attention parameters.

3.3 ADAPTIVE TEST-TIME SAMPLING FOR TRANSOLVER

We next construct the inference-time sampling measure from geometry, a draft prediction, or intermediate backbone features.

Let $r_\psi : \Omega \rightarrow [0, \infty)$ be a nonnegative refinement score, where ψ denotes the information used to construct it, and define

$$\nu_\psi(x) := \frac{r_\psi(x)}{\int_{\Omega} r_\psi(y) d\rho(y)}, \quad \tilde{\rho}_{\gamma,\psi} := (1 - \gamma)\rho + \gamma(\nu_\psi * \rho), \quad \gamma \in [0, 1]. \quad (14)$$

Here $\nu_\psi * \rho$ denotes the reweighted empirical measure induced by the refinement score. The correction used in the point-to-slice aggregation is

$$c_{\gamma,\psi}(x) := \frac{d\rho}{d\tilde{\rho}_{\gamma,\psi}}(x) = \frac{1}{(1 - \gamma) + \gamma\nu_\psi(x)}. \quad (15)$$

3.3.1 PHYSICS-INFORMED DENSITY REFINEMENT

The first policy constructs r_ψ from prior information about the geometry or the governing physical problem. Let $s_{\text{prior}}(x) \geq 0$ be a problem-dependent indicator whose large values identify regions that are expected to require finer spatial resolution. This indicator can be obtained from a known geometric feature, a numerical refinement rule, or a physical quantity available before inference. We set

$$\nu_{\text{prior}}(x) := \frac{s_{\text{prior}}(x)}{\int_{\Omega} s_{\text{prior}}(y) d\rho(y)} \quad (16)$$

and use it in equation 14 to sample the final point set.

This policy does not require a preliminary network evaluation; once the prior indicator is available, points are sampled from $\tilde{\rho}_{\gamma,\text{prior}}$ and inferred with the correction in equation 15.

3.3.2 COARSE-PREDICTION-GUIDED DENSITY REFINEMENT

This strategy follows the test-time refinement perspective of PDE-Refiner (Lippe et al., 2023) and the initial-prediction–difficulty-estimation pipeline of ATCA (Jenkins, 2026). When no reliable prior is available, we draw a small coarse set S_0 from ρ and evaluate the frozen operator on this set:

$$\hat{u}^{(0)} := \mathcal{G}_\theta(a|_{S_0}), \quad |S_0| \ll N, \quad (17)$$

where $a|_{S_0}$ denotes the input field restricted to the coarse points. From $\hat{u}^{(0)}$, we compute a non-negative local-variation score, for example by combining discrete estimates of the gradient and Laplacian:

$$s_{\text{coarse}}(x) := \Phi \left(\left\| \nabla \hat{u}^{(0)}(x) \right\|_2, \left\| \Delta \hat{u}^{(0)}(x) \right\|_2 \right), \quad \Phi \geq 0, \quad (18)$$

where Φ is a fixed nonnegative combination and the derivatives are evaluated with the available mesh or point-cloud neighborhood. The score is transferred to candidate points, normalized as in equation 14, and used to sample $\tilde{\rho}_{\gamma, \text{coarse}}$.

The method works because the draft variation score directs the final spatial resolution toward regions where the field is changing rapidly.

3.3.3 CACHE-AWARE LEARNED REFINEMENT

The coarse prediction provides features correlated with local physical or geometric variation, but the subsequent computation of the first pass is otherwise discarded. We reuse it as follows. Let $S_1 = S_0 \cup B$ be the final point set, where B contains the newly allocated points. For the first Physics-Attention layer, let h_i^{in} be its pointwise input feature and $w_{i,j}^{(1)}$ its slice weight. Define

$$A_j(S) := \sum_{i \in S} c_i w_{i,j}^{(1)} h_i^{\text{in}}, \quad D_j(S) := \sum_{i \in S} c_i w_{i,j}^{(1)}. \quad (19)$$

Because the summands are pointwise, their values on S_0 are unchanged after adding B , and

$$A_j(S_1) = A_j(S_0) + A_j(B), \quad D_j(S_1) = D_j(S_0) + D_j(B). \quad (20)$$

so the coarse values can be cached and only the contributions from B need to be computed. We train a lightweight head H_ϕ with the frozen backbone, attach it to the first-block feature $h_i^{(1)}$, and define

$$\hat{u}_i^{\text{head}} = H_\phi(h_i^{(1)}), \quad s_{\text{head}}(x_i) := \Phi \left(\left\| \nabla \hat{u}^{\text{head}}(x_i) \right\|_2, \left\| \Delta \hat{u}^{\text{head}}(x_i) \right\|_2 \right), \quad (21)$$

and use this score in equation 14. Token attention, deslicing, and later blocks still run on S_1 , because B changes the global token states.

3.4 DISTRIBUTION-AMORTIZED TRAINING

Using the measure-consistency property above, we extend the geometry-amortized training strategy of Transolver-3 (Zhou et al., 2026), which trains on random subsets of a high-resolution mesh, to distribution-amortized training. At step t , we draw a point count N_t from a prescribed range and choose a refinement configuration (γ_t, ψ_t) , giving

$$\rho_t := \tilde{\rho}_{\gamma_t, \psi_t}, \quad x_i^{(t)} \sim \rho_t, \quad i = 1, \dots, N_t, \quad S_t := \{x_i^{(t)}\}_{i=1}^{N_t}. \quad (22)$$

The configuration (γ_t, ψ_t) may be random or generated from a geometry- or physics-based refinement rule. We use the point-to-slice correction from equation 15 during the forward pass and optimize

$$\min_\theta \mathbb{E}_t \left[\ell \left(\mathcal{G}_\theta(a|_{S_t}), u|_{S_t} \right) \right]. \quad (23)$$

Because each corrected point-to-slice aggregation remains an estimator of the same ρ -based operator, changing N_t and ρ_t across training steps provides distributional augmentation without changing the operator target. Uniform variable-cardinality training is recovered when $\gamma_t = 0$.

4 EXPERIMENTS

We evaluate AdaSolver on five standard PDE benchmarks and three industrial design prediction tasks, covering both surface and volumetric fields on three-dimensional geometries. We further assess its efficiency in two CFD-based geometry optimization tasks involving two-dimensional airfoils and three-dimensional wings.

4.1 BENCHMARKS

Our evaluation spans canonical PDE problems and representative industrial applications. The standard PDE suite comprises Elasticity, Plasticity, Airfoil, Pipe (Li et al., 2023b) and Darcy flow (Li et al., 2021). These public datasets are widely used to benchmark neural operators (Hao et al., 2023; Wu et al., 2024).

For industrial applications, we consider ShapeNet-Car (Umetani & Bickel, 2018) and DrivAerML (Ashton et al., 2024) for automotive aerodynamics, and SuperWing (Yang et al., 2025) for three-dimensional wing design. In addition to physical-field prediction errors, we evaluate the accuracy of design-relevant aerodynamic quantities, including the drag coefficient C_D , lift coefficient C_L , and drag-to-lift ratio C_D/C_L . We also use the learned PDE surrogates to guide shape optimization of a two-dimensional airfoil and a three-dimensional wing, assessing computational efficiency and aerodynamic prediction accuracy during optimization.

4.2 IMPLEMENTATION AND EVALUATION

As described in equation 3 and equation 4, Transolver approximates a continuous integral operator through point aggregation and attention over latent physical states. With the number of physical states fixed, varying the spatial sampling budget changes the number of point samples used to estimate the aggregation integrals. For Airfoil, Pipe, Elasticity, and Plasticity, we adopt geometry-amortized training from Transolver-3 (Zhou et al., 2026). Given a full mesh of N points, we draw a point count N_t uniformly from $\{\text{round}(0.4N), \dots, N\}$ for each minibatch, then sample N_t mesh points uniformly without replacement. Darcy uses the existing backbone trained on full meshes. The trained backbone is frozen when evaluating different sampling policies and inference budgets.

For a consistent comparison, we treat all input discretizations as unstructured meshes and evaluate sampling budgets of 37.5%, 50%, and 75% of the full-mesh point count. Uniform subsampling of mesh points serves as the common baseline. We report relative L_2 errors for physical-field predictions and summarize performance across budgets using the relative area under the error–budget curve (rAUEC). Let b_k denote the retained fraction of mesh points, with $(b_1, b_2, b_3) = (0.375, 0.50, 0.75)$, and let $E_{\text{method}}(b)$ and $E_{\text{uniform}}(b)$ denote mean test errors for a given sampling seed. Using trapezoidal integration, we compute

$$\text{rAUEC} := \frac{\sum_{k=1}^2 (b_{k+1} - b_k) [E_{\text{method}}(b_k) + E_{\text{method}}(b_{k+1})]}{\sum_{k=1}^2 (b_{k+1} - b_k) [E_{\text{uniform}}(b_k) + E_{\text{uniform}}(b_{k+1})]}. \quad (24)$$

Lower rAUEC is better, with uniform sampling normalized to 1. We report mean rAUEC and its sample standard deviation over three independent sampling runs with matched Uniform baselines. We also report end-to-end runtime for geometry optimization.

Airfoil uses the same 112 cases with reference $C_L > 0.01$ for field and C_D/C_L evaluation, with metric-specific sampling policies and absolute error for C_D/C_L . Elasticity uses 200 independent holdout cases.

For industrial benchmarks, ShapeNet-Car combines full-cloud velocity relative L_2 with half the surface-pressure relative L_2 error. DrivAerML uses surface-pressure relative L_2 , and SuperWing uses joint relative L_2 on the dataset-scaled pressure and skin-friction channels. The aerodynamic metrics are relative errors in C_D for the car datasets and C_D/C_L for SuperWing, with sampling policies selected separately for each metric. ShapeNet-Car and DrivAerML use 100 and 50 test geometries, respectively. SuperWing uses 2,871 cases for field evaluation and a fixed 2,563-case subset for C_D/C_L , selected by a calibration-derived lift-magnitude cutoff of approximately 0.071.

4.3 MAIN RESULTS

Standard Benchmarks. Table 1 compares physics-informed density refinement (PI), coarse-prediction-guided density refinement (CPG), and cache-aware learned refinement (CA) with uniform subsampling across five standard PDE benchmarks. Overall, adaptive sampling reduces rAUEC, showing that point density is a design variable for improving neural operator predictions under a fixed spatial budget. CPG generally improves upon PI, indicating that features of the predicted solution can guide point allocation more effectively than geometry and prescribed heuristics alone. CA achieves performance close to CPG on several benchmarks, supporting the learnability of the field-dependent signals used for density refinement. The differences are particularly clear on Darcy: PI provides little improvement over uniform sampling, whereas CPG and CA achieve similar, substantially lower rAUEC. Thus, even on a regular Cartesian grid, solution-informed density allocation can improve prediction accuracy. These results suggest that the benefit of adaptive sampling depends on how well its allocation signal reflects the solution field, beyond the geometric complexity of the domain.

Table 1: rAUEC ($\downarrow$) on standard PDE benchmarks over three seeds. Means appear above standard deviations. Best and second-best means are **bold** and underlined, respectively.

Method	Airfoil2D		Pipe	Darcy	Elasticity	Plasticity
	Field	C_D/C_L	Field	Field	Field	Field
PI	0.6177 ± 0.0187	0.8010 ± 0.2108	0.8711 ± 0.0061	0.9965 ± 0.0177	0.7280 ± 0.0083	0.8888 ± 0.0041
CPG	<u>0.6739</u> ± 0.0232	<u>0.4907</u> ± 0.3151	0.6719 ± 0.0098	<u>0.6419</u> ± 0.0049	0.4681 ± 0.0041	<u>0.8890</u> ± 0.0038
CA	0.6888 ± 0.0119	0.4128 ± 0.2384	<u>0.7051</u> ± 0.0116	0.6400 ± 0.0062	<u>0.5469</u> ± 0.0021	0.8897 ± 0.0049

Industrial Applications. Wind-tunnel testing informs vehicle and aircraft design by assessing aerodynamic performance through quantities such as drag and lift coefficients. Table 2 reports both full-field prediction errors and errors in aerodynamic quantities on ShapeNet-Car, DrivAerML, and SuperWing. The results show that all three methods substantially improve neural operator accuracy across sampling budgets. For aerodynamic quantities, concentrating samples in influential regions can match or even surpass full-mesh accuracy with only 37.5% of the points. This agrees with the physical intuition that some flow regions contribute more strongly to aerodynamic forces than others. Accurate predictions with fewer points can accelerate geometry optimization.

Table 2: rAUEC ($\downarrow$) on industrial benchmarks over three seeds. Means appear above standard deviations. Best and second-best means are **bold** and underlined, respectively.

Method	ShapeNet-Car		DrivAerML		SuperWing	
	Field	C_D	Field	C_D	Field	C_D/C_L
PI	0.58952 ± 0.00180	0.65606 ± 0.00168	0.88342 ± 0.00011	0.92455 ± 0.00149	0.75640 ± 0.00079	0.82456 ± 0.00539
CPG	0.55770 ± 0.00219	0.64743 ± 0.00454	0.82458 ± 0.00005	0.92093 ± 0.00148	0.67059 ± 0.00212	0.80259 ± 0.00761
CA	<u>0.55774</u> ± 0.00222	<u>0.64786</u> ± 0.00404	<u>0.82635</u> ± 0.00019	<u>0.92438</u> ± 0.00153	<u>0.71927</u> ± 0.00211	<u>0.81675</u> ± 0.00700

4.4 INVERSE DESIGN

We use the trained AdaSolver surrogate with the sampling policies tailored to C_D/C_L described above to optimize airfoil and wing geometries. The design objective is to minimize the drag-to-lift ratio.

2D Airfoil. We adopt the airfoil design case from Geo-FNO (Li et al., 2023b). Figure 3 compares optimization using AdaSolver with CA sampling against optimization using the same surrogate on the full mesh. AdaSolver substantially accelerates the optimization process while maintaining comparable final aerodynamic performance, as verified by CFD, and reaches better designs within the same time budget. The speedup comes from reducing the cost of repeated objective and gradient evaluations through adaptive point allocation.

5 CONCLUSION

In this paper, we propose AdaSolver, a measure-aware framework for adaptive spatial computation in point-based neural operators. By correcting sampling-induced bias in latent aggregation, AdaSolver supports adaptive point allocation and distribution-amortized training while remaining consistent with the reference operator. Experiments on five standard PDE benchmarks, three industrial aerodynamic prediction tasks, and airfoil and wing optimization demonstrate improved accuracy at matched point budgets and faster surrogate-based design. This study establishes spatial sampling as an axis of test-time scaling for point-based neural operators, offering a direction for more efficient simulation and design in science and engineering.

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A SAMPLING-MEASURE BIAS IN PHYSICS-ATTENTION

We analyze the effect of evaluating a frozen Physics-Attention layer under a sampling measure different from the one used during training. Let (Ω, ρ) be the physical domain equipped with the reference probability measure, let $h : \Omega \rightarrow \mathbb{R}^C$ be the point-feature field, and let Ω_s be a continuous slice domain of finite measure. A bounded measurable slice-weight function $w : \Omega \times \Omega_s \rightarrow [0, \infty)$ satisfies

$$\int_{\Omega_s} w(x, \xi_s) d\xi_s = 1, \quad \int_{\Omega} w(x, \xi_s) d\rho(x) > 0. \quad (\text{A.1})$$

For each $\xi_s \in \Omega_s$, define the slice-weighted probability measure

$$d\pi_{\xi_s}(x) := \frac{w(x, \xi_s)}{\int_{\Omega} w(y, \xi_s) d\rho(y)} d\rho(x). \quad (\text{A.2})$$

Let $\tilde{\rho}$ be the inference-time sampling measure. We assume $\tilde{\rho} \ll \rho$ and write

$$\lambda := \frac{d\tilde{\rho}}{d\rho} > 0 \quad \rho\text{-a.e.} \quad (\text{A.3})$$

The reference and shifted physics-token fields are

$$u_s^\rho(\xi_s) := \mathbb{E}_{\pi_{\xi_s}}[h], \quad u_s^{\tilde{\rho}}(\xi_s) := \frac{\mathbb{E}_{\pi_{\xi_s}}[\lambda h]}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]}. \quad (\text{A.4})$$

Since $\lambda > 0$ holds ρ -almost everywhere and w is bounded with positive ρ -mass,

$$\mathbb{E}_{\pi_{\xi_s}}[\lambda] = \frac{\int_{\Omega} w(x, \xi_s) d\tilde{\rho}(x)}{\int_{\Omega} w(x, \xi_s) d\rho(x)} \in (0, \infty).$$

Hence, $u_s^{\tilde{\rho}}$ is well defined, and its discrepancy from u_s^ρ reflects a change in the limiting tokenization operator rather than finite-sample quadrature error.

Proposition A.1 (Exact bias of a physics token). *Fix $\xi_s \in \Omega_s$. Assume $h \in L^2(\pi_{\xi_s}; \mathbb{R}^C)$ and $\lambda h \in L^1(\pi_{\xi_s}; \mathbb{R}^C)$. Then*

$$u_s^{\tilde{\rho}}(\xi_s) - u_s^\rho(\xi_s) = \frac{\text{Cov}_{\pi_{\xi_s}}(\lambda, h)}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]} = \mathbb{E}_{\pi_{\xi_s}} \left[\left(\frac{\lambda}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]} - 1 \right) h \right], \quad (\text{A.5})$$

where

$$\text{Cov}_{\pi_{\xi_s}}(\lambda, h) := \mathbb{E}_{\pi_{\xi_s}}[(\lambda - \mathbb{E}_{\pi_{\xi_s}}[\lambda])(h - \mathbb{E}_{\pi_{\xi_s}}[h])] \in \mathbb{R}^C. \quad (\text{A.6})$$

Furthermore, if $\lambda \in L^2(\pi_{\xi_s})$, define the bounded linear operators from $L^2(\pi_{\xi_s}; \mathbb{R}^C)$ to $\mathbb{R}^C$ by

$$\mathcal{P}_{\xi_s}^\rho f := \mathbb{E}_{\pi_{\xi_s}}[f], \quad \mathcal{P}_{\xi_s}^{\tilde{\rho}} f := \frac{\mathbb{E}_{\pi_{\xi_s}}[\lambda f]}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]}. \quad (\text{A.7})$$

Their operator-norm discrepancy is exactly

$$\|\mathcal{P}_{\xi_s}^{\tilde{\rho}} - \mathcal{P}_{\xi_s}^\rho\|_{\text{op}} = \left\| \frac{\lambda}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]} - 1 \right\|_{L^2(\pi_{\xi_s})} = \frac{\sqrt{\text{Var}_{\pi_{\xi_s}}(\lambda)}}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]}. \quad (\text{A.8})$$

Proof. By equation A.4,

$$u_s^{\tilde{\rho}}(\xi_s) - u_s^{\rho}(\xi_s) = \frac{\mathbb{E}_{\pi_{\xi_s}}[\lambda h] - \mathbb{E}_{\pi_{\xi_s}}[\lambda] \mathbb{E}_{\pi_{\xi_s}}[h]}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]} = \frac{\text{Cov}_{\pi_{\xi_s}}(\lambda, h)}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]}, \quad (\text{A.9})$$

which also equals the second expression in equation A.5.

For the operator statement, set

$$r_{\xi_s} := \frac{\lambda}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]} - 1. \quad (\text{A.10})$$

Then, for every $f \in L^2(\pi_{\xi_s}; \mathbb{R}^C)$,

$$(\mathcal{P}_{\xi_s}^{\tilde{\rho}} - \mathcal{P}_{\xi_s}^{\rho})f = \mathbb{E}_{\pi_{\xi_s}}[r_{\xi_s} f]. \quad (\text{A.11})$$

Cauchy–Schwarz gives

$$\|\mathbb{E}_{\pi_{\xi_s}}[r_{\xi_s} f]\|_2 \leq \|r_{\xi_s}\|_{L^2(\pi_{\xi_s})} \|f\|_{L^2(\pi_{\xi_s}; \mathbb{R}^C)}, \quad (\text{A.12})$$

so the operator norm is at most $\|r_{\xi_s}\|_{L^2(\pi_{\xi_s})}$. If $r_{\xi_s} \neq 0$, choose a unit vector $e \in \mathbb{R}^C$ and

$$f^*(x) := \frac{r_{\xi_s}(x)}{\|r_{\xi_s}\|_{L^2(\pi_{\xi_s})}} e. \quad (\text{A.13})$$

Then $\|f^*\|_{L^2(\pi_{\xi_s}; \mathbb{R}^C)} = 1$ and

$$(\mathcal{P}_{\xi_s}^{\tilde{\rho}} - \mathcal{P}_{\xi_s}^{\rho})f^* = \|r_{\xi_s}\|_{L^2(\pi_{\xi_s})} e, \quad (\text{A.14})$$

which attains the upper bound. The case $r_{\xi_s} = 0$ is immediate. Finally,

$$\|r_{\xi_s}\|_{L^2(\pi_{\xi_s})}^2 = \frac{\mathbb{E}_{\pi_{\xi_s}}[(\lambda - \mathbb{E}_{\pi_{\xi_s}}[\lambda])^2]}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]^2} = \frac{\text{Var}_{\pi_{\xi_s}}(\lambda)}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]^2}, \quad (\text{A.15})$$

which proves equation A.8. $\square$

A.1 PROPAGATION THROUGH CONTINUOUS PHYSICS-ATTENTION

We first recall the continuous formulation leading to equation 13 of Transolver (Wu et al., 2024). Let $g : \Omega \rightarrow \Omega_s$ be a diffeomorphism from the physical domain to the slice domain, with

$$J_g(\xi_s) := |\det(\nabla_{\xi_s} g^{-1}(\xi_s))|. \quad (\text{A.16})$$

For a bounded physics-token field $u_s : \Omega_s \rightarrow \mathbb{R}^C$, the slice-to-slice Physics-Attention kernel is

$$\begin{aligned} \ell_{u_s}(\xi'_s, \xi_s) &:= \frac{(W_q u_s(\xi'_s))^\top W_k u_s(\xi_s)}{\tau}, \\ \alpha_{u_s}(\xi'_s, \xi_s) &:= \frac{\exp(\ell_{u_s}(\xi'_s, \xi_s))}{\int_{\Omega_s} \exp(\ell_{u_s}(\xi'_s, \zeta_s)) d\zeta_s}, \\ \kappa_{ss, u_s}(\xi'_s, \xi_s) &:= \alpha_{u_s}(\xi'_s, \xi_s) W_v, \end{aligned} \quad (\text{A.17})$$

where $W_q, W_k \in \mathbb{R}^{d \times C}$, $W_v \in \mathbb{R}^{C \times C}$, and $\tau > 0$ are fixed. Thus, κ_{ss, u_s} is the matrix-valued kernel that performs attention among physics tokens. After expressing the physical-to-slice kernel as a weighted combination of κ_{ss, u_s} , the continuous Physics-Attention operator becomes

$$\begin{aligned} \mathcal{T}[u_s](x) &:= \int_{\Omega_s} w(x, \xi'_s) \int_{\Omega_s} \kappa_{ss, u_s}(\xi'_s, \xi_s) u_s(\xi_s) J_g(\xi_s) d\xi_s d\xi'_s \\ &= \int_{\Omega_s} w(x, \xi'_s) \int_{\Omega_s} \alpha_{u_s}(\xi'_s, \xi_s) W_v u_s(\xi_s) J_g(\xi_s) d\xi_s d\xi'_s. \end{aligned} \quad (\text{A.18})$$

This is the deslice–attention–token factorization in the continuous derivation of Transolver. If the slice domain is represented by M tokens $s_1, \dots, s_M$, with $q_j = W_q s_j$, $k_j = W_k s_j$, and $v_j = W_v s_j$, its Monte Carlo discretization is

$$h'_i = \sum_{j=1}^M w_{i,j} \sum_{t=1}^M \frac{\exp(q_j^\top k_t / \tau)}{\sum_{p=1}^M \exp(q_j^\top k_p / \tau)} v_t, \quad i = 1, \dots, N, \quad (\text{A.19})$$

which is equation 13 of Transolver under the simplifying choice $J_g \equiv 1$. For the token fields in equation A.4, we write $\mathcal{G}_\rho(h) := \mathcal{T}[u_s^\rho]$ and $\mathcal{G}_{\tilde{\rho}}(h) := \mathcal{T}[u_s^{\tilde{\rho}}]$. The map g , the slice weight w , and the projection matrices are held fixed when the sampling measure changes.

Lemma A.2 (Exact output perturbation for a token shift). *Let $u_s, \delta_s \in L^\infty(\Omega_s; \mathbb{R}^C)$. The logit perturbation induced by replacing u_s with $u_s + \delta_s$ is*

$$\begin{aligned} \Delta \ell_{\delta_s}(\xi'_s, \xi_s) &:= \ell_{u_s + \delta_s}(\xi'_s, \xi_s) - \ell_{u_s}(\xi'_s, \xi_s) \\ &= \frac{(W_q \delta_s(\xi'_s))^\top W_k u_s(\xi_s) + (W_q u_s(\xi'_s))^\top W_k \delta_s(\xi_s) + (W_q \delta_s(\xi'_s))^\top W_k \delta_s(\xi_s)}{\tau}. \end{aligned} \quad (\text{A.20})$$

For each query ξ'_s , define

$$R_{\delta_s}(\xi'_s, \xi_s) := \frac{\exp(\Delta \ell_{\delta_s}(\xi'_s, \xi_s))}{\int_{\Omega_s} \alpha_{u_s}(\xi'_s, \zeta_s) \exp(\Delta \ell_{\delta_s}(\xi'_s, \zeta_s)) d\zeta_s}. \quad (\text{A.21})$$

Then $\alpha_{u_s + \delta_s} = \alpha_{u_s} R_{\delta_s}$, and

$$\begin{aligned} \mathcal{T}[u_s + \delta_s](x) - \mathcal{T}[u_s](x) &= \int_{\Omega_s} w(x, \xi'_s) \int_{\Omega_s} \alpha_{u_s}(\xi'_s, \xi_s) \left[(R_{\delta_s}(\xi'_s, \xi_s) - 1) W_v u_s(\xi_s) \right. \\ &\quad \left. + R_{\delta_s}(\xi'_s, \xi_s) W_v \delta_s(\xi_s) \right] J_g(\xi_s) d\xi_s d\xi'_s. \end{aligned} \quad (\text{A.22})$$

The two terms are the attention-redistribution bias and the value-token bias, respectively.

Proof. By equation A.20,

$$\begin{aligned} \alpha_{u_s + \delta_s}(\xi'_s, \xi_s) &= \frac{\exp(\ell_{u_s}(\xi'_s, \xi_s)) \exp(\Delta \ell_{\delta_s}(\xi'_s, \xi_s))}{\int_{\Omega_s} \exp(\ell_{u_s}(\xi'_s, \zeta_s)) \exp(\Delta \ell_{\delta_s}(\xi'_s, \zeta_s)) d\zeta_s} \\ &= \alpha_{u_s}(\xi'_s, \xi_s) R_{\delta_s}(\xi'_s, \xi_s). \end{aligned} \quad (\text{A.23})$$

Substituting this identity and $u_s + \delta_s$ into equation A.18, then subtracting $\mathcal{T}[u_s]$, gives equation A.22. $\square$

Theorem A.3 (Exact propagation of sampling-measure bias). *Assume the conditions of Proposition A.1 hold for every $\xi_s \in \Omega_s$, and define the token perturbation*

$$\delta_s^{\tilde{\rho}}(\xi_s) := u_s^{\tilde{\rho}}(\xi_s) - u_s^{\rho}(\xi_s) = \frac{\text{Cov}_{\pi_{\xi_s}}(\lambda, h)}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]}. \quad (\text{A.24})$$

If $u_s^{\rho}, \delta_s^{\tilde{\rho}} \in L^\infty(\Omega_s; \mathbb{R}^C)$, then

$$\begin{aligned} \mathcal{G}_{\tilde{\rho}}(h)(x) - \mathcal{G}_{\rho}(h)(x) &= \int_{\Omega_s} w(x, \xi'_s) \int_{\Omega_s} \alpha_{u_s^{\rho}}(\xi'_s, \xi_s) \left[(R_{\delta_s^{\tilde{\rho}}}(\xi'_s, \xi_s) - 1) W_v u_s^{\rho}(\xi_s) \right. \\ &\quad \left. + R_{\delta_s^{\tilde{\rho}}}(\xi'_s, \xi_s) W_v \frac{\text{Cov}_{\pi_{\xi_s}}(\lambda, h)}{\mathbb{E}_{\pi_{\xi_s}}[\lambda]} \right] J_g(\xi_s) d\xi_s d\xi'_s, \end{aligned} \quad (\text{A.25})$$

where $R_{\delta_s^{\tilde{\rho}}}$ is given by equation A.21 with base field $u_s = u_s^{\rho}$. Thus, the exact output bias is the sum of the change in attention weights and the direct propagation of the biased value tokens.

Proof. Equation A.24 follows from Proposition A.1. Applying Lemma A.2 with $u_s = u_s^{\rho}$ and $\delta_s = \delta_s^{\tilde{\rho}}$ gives equation A.25. $\square$

We next consider a small change of sampling measure,

$$d\rho_\varepsilon(x) = (1 + \varepsilon \eta(x)) d\rho(x), \quad \mathbb{E}_\rho[\eta] = 0, \quad (\text{A.26})$$

where $\eta \in L^\infty(\rho)$ and $|\varepsilon|$ is small enough that $1 + \varepsilon \eta > 0$. Let $u_s^{\rho_\varepsilon}$ be the token field under ρ_ε . Proposition A.1 gives

$$u_s^{\rho_\varepsilon}(\xi_s) - u_s^{\rho}(\xi_s) = \frac{\varepsilon \text{Cov}_{\pi_{\xi_s}}(\eta, h)}{1 + \varepsilon \mathbb{E}_{\pi_{\xi_s}}[\eta]} = \varepsilon \text{Cov}_{\pi_{\xi_s}}(\eta, h) + \mathcal{O}(\varepsilon^2), \quad (\text{A.27})$$

uniformly in ξ_s . The corresponding first-order change of the attention logit is

$$\begin{aligned} \ell_\eta^{(1)}(\xi'_s, \xi_s) &:= \frac{1}{\tau} \left[(W_q \text{Cov}_{\pi_{\xi_s}}(\eta, h))^\top W_k u_s^{\rho}(\xi_s) \right. \\ &\quad \left. + (W_q u_s^{\rho}(\xi'_s))^\top W_k \text{Cov}_{\pi_{\xi_s}}(\eta, h) \right]. \end{aligned} \quad (\text{A.28})$$

Theorem A.4 (First-order sampling-measure bias). Assume $h \in L^\infty(\rho; \mathbb{R}^C)$, $J_g \in L^\infty(\Omega_s)$, and the preceding conditions on w and η . Then, in $L^\infty(\Omega; \mathbb{R}^C)$,

$$\mathcal{G}_{\rho_\varepsilon}(h) - \mathcal{G}_\rho(h) = \varepsilon \mathcal{S}_{\rho,h}[\eta] + \mathcal{O}(\varepsilon^2), \quad (\text{A.29})$$

where

$$\begin{aligned} \mathcal{S}_{\rho,h}[\eta](x) = & \int_{\Omega_s} w(x, \xi'_s) \int_{\Omega_s} \alpha_{u_s^\rho}(\xi'_s, \xi_s) \left\{ W_v \text{Cov}_{\pi_{\xi_s}}(\eta, h) \right. \\ & \left. + \left[\ell_\eta^{(1)}(\xi'_s, \xi_s) - \int_{\Omega_s} \alpha_{u_s^\rho}(\xi'_s, \zeta_s) \ell_\eta^{(1)}(\xi'_s, \zeta_s) d\zeta_s \right] W_v u_s^\rho(\xi_s) \right\} J_g(\xi_s) d\xi_s d\xi'_s. \end{aligned} \quad (\text{A.30})$$

Consequently,

$$\lim_{\varepsilon \rightarrow 0} \frac{\|\mathcal{G}_{\rho_\varepsilon}(h) - \mathcal{G}_\rho(h)\|_{L^\infty(\Omega)}}{|\varepsilon|} = \|\mathcal{S}_{\rho,h}[\eta]\|_{L^\infty(\Omega)}. \quad (\text{A.31})$$

If $\mathcal{S}_{\rho,h}[\eta] \neq 0$, then the output bias is $\Theta(|\varepsilon|)$.

Proof. The boundedness assumptions make the remainder in equation A.27 uniform over Ω_s . By bilinearity of the query-key logit, equation A.20 therefore gives

$$\Delta \ell_{u_s^{\rho_\varepsilon} - u_s^\rho}(\xi'_s, \xi_s) = \varepsilon \ell_\eta^{(1)}(\xi'_s, \xi_s) + \mathcal{O}(\varepsilon^2) \quad (\text{A.32})$$

uniformly on $\Omega_s \times \Omega_s$. Expanding the numerator and denominator of equation A.21 yields

$$\begin{aligned} R_{u_s^{\rho_\varepsilon} - u_s^\rho}(\xi'_s, \xi_s) = & 1 + \varepsilon \left[\ell_\eta^{(1)}(\xi'_s, \xi_s) \right. \\ & \left. - \int_{\Omega_s} \alpha_{u_s^\rho}(\xi'_s, \zeta_s) \ell_\eta^{(1)}(\xi'_s, \zeta_s) d\zeta_s \right] + \mathcal{O}(\varepsilon^2), \end{aligned} \quad (\text{A.33})$$

also uniformly. Moreover, equation A.27 implies

$$W_v(u_s^{\rho_\varepsilon}(\xi_s) - u_s^\rho(\xi_s)) = \varepsilon W_v \text{Cov}_{\pi_{\xi_s}}(\eta, h) + \mathcal{O}(\varepsilon^2). \quad (\text{A.34})$$

Substituting equation A.33 and equation A.34 into Lemma A.2 and collecting the coefficient of ε gives equation A.30. Since J_g is bounded and $\int_{\Omega_s} w(x, \xi'_s) d\xi'_s = 1$, the integrated remainder is $\mathcal{O}(\varepsilon^2)$ in $L^\infty(\Omega)$, proving equation A.29.

Finally, the reverse triangle inequality gives

$$\left| \frac{\|\mathcal{G}_{\rho_\varepsilon}(h) - \mathcal{G}_\rho(h)\|_{L^\infty(\Omega)}}{|\varepsilon|} - \|\mathcal{S}_{\rho,h}[\eta]\|_{L^\infty(\Omega)} \right| = \mathcal{O}(|\varepsilon|), \quad (\text{A.35})$$

which proves equation A.31 and the stated $\Theta(|\varepsilon|)$ order. $\square$

The Jacobian J_g remains in every bias term. Setting $J_g \equiv 1$ recovers the simplifying assumption used by Transolver when deriving its discrete Physics-Attention update.